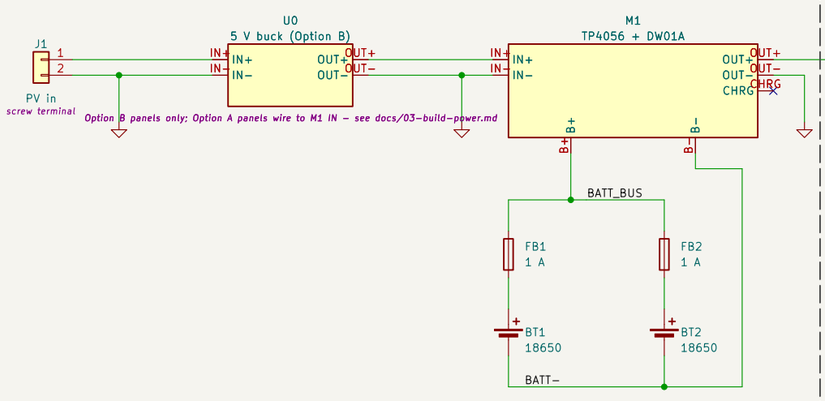
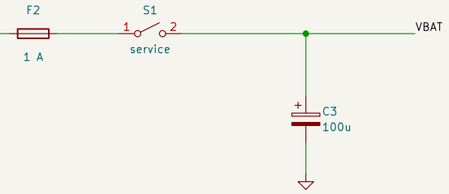
1 Solar input, charger & cells



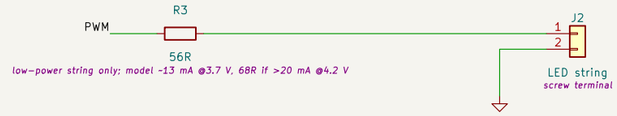
1S2P − each cell fused off the charger B+ node. TP4056 CHRG not wired.

2 Battery rail − fuse + service switch



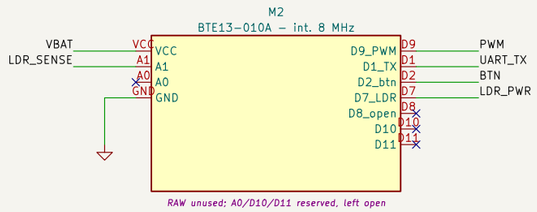
VBAT feeds the Pro Mini, U3 and R5 by net label

3 LED output − direct pin drive

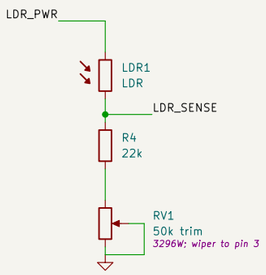


No driver transistor: D9 PWM drives the low-power string via R3

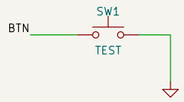
4 Controller − Pro Mini BTE13-010A



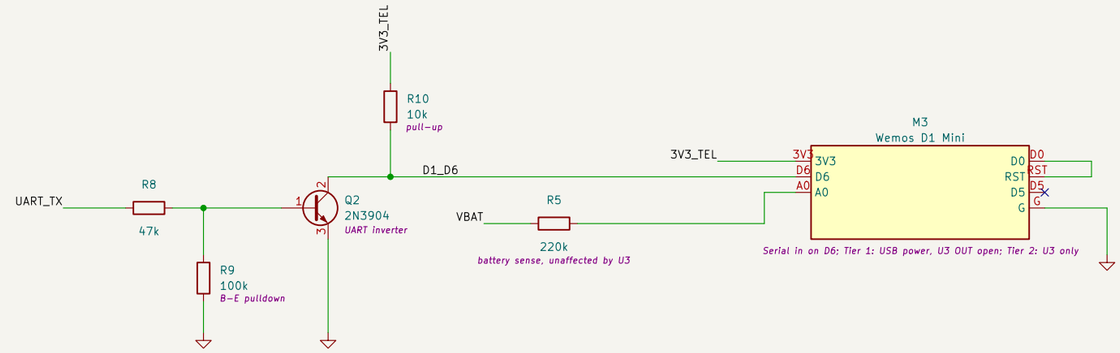
Light sensor (switched by D7), RV1 sets dusk



Test button

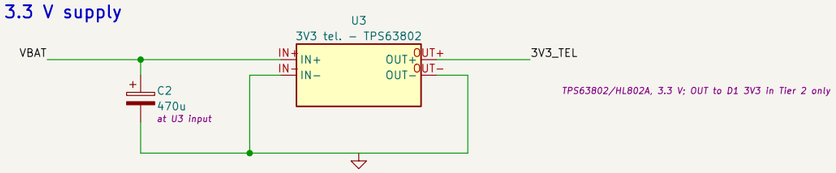


5 Telemetry − Wemos D1 Mini



Arduino D1/TX −> inverted 9600-baud UART −> D6. D0-RST for deep-sleep wake.

6 Telemetry 3.3 V supply



Flow: panel −> U0 −> TP4056 −> battery rail −> Pro Mini,
direct PWM drive of a low-power LED string, D1 Mini telemetry
Controller D1/TX −> R8/Q2/R10 −> D1 Mini D6 (inverted 9600-baud serial).
Tier 1: D1 Mini on USB, U3 OUT open. Tier 2: U3 OUT −> 3V3, D0 −> RST.

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Title: Solar Lights Lite

Size: A3	Date: 30-Sep-2026	**Rev: 0.6**
KiCad E.D.A. 10.0.6		Id: 1/1